BUILT WITH CLAUDE AND THREE.JS

HOW TO BUILD A HOUSE THAT BUILDS ITSELF ON SCROLL



THE FULL BLUEPRINT

EVERY PROMPT AND EVERY FILE,
FROM ONE PHOTO TO THE CLIENT

OMNYA LAB

START HERE

EVERYTHING
IN THIS GUIDE

This is the whole method behind our Hold to build house, the one that goes from a blueprint on the desk to a finished home while you scroll or hold the screen, with the camera turning all the way around it. You can see it live at **omnyalab.com/demo/hold-to-build**, and every step, prompt, number and file in these pages comes from that house.



PHASE 04 OF 09, THE FULL FRAME

WHAT YOU NEED

- **Claude Code** and **three.js**, both free, for the 3D and the page
- **Nano Banana Pro** on Higgsfield for the photos
- **Kling 4.0** on Higgsfield for the clips
- **ffmpeg**, free, for the frames
- A free **Vercel** account to put it online

WHAT IT COST US

- Nine clips on Kling 4.0 at 1080p, 5 seconds each, **16.25 credits a clip**
- Two video edits for the walk inside, **40 credits each**

- Photos on Nano Banana Pro, about ten with the retakes
- No 3D software, no render farm, no film crew

THE CHAPTERS

00 **The fast way: hand it to Claude Code**

01 **The photo that sets the house**

02 **The house in code**

03 **The rail and the screenshots**

04 **The phases, backwards**

05 **Nine photos, camera locked**

06 **Nine clips, start and end**

07 **Frames light enough for a phone**

08 **The hold to build page**

09 **Delivery and what to charge**



CHAPTER 00

THE FAST WAY

Put this PDF in an empty folder, open Claude Code in it and paste the prompt below. Claude reads the guide and walks you through it one step at a time, waiting for your OK after each one.

PROMPT 00 · CLAUDE CODE · THE WHOLE METHOD

```
I bought the Full Build Blueprint "How to build a house that builds
itself" by Omnya Lab. The PDF is in this folder: read it page by
page, from chapter 01 to chapter 09, before you do anything.

I want what Omnya Lab did with their Hold to build house
(omnyalab.com/demo/hold-to-build): one photo of a finished house, the
same house copied in three.js, a camera that turns around it on one
rail, one photo per phase of the build with the camera locked on the
3D, one clip per phase, and a page that plays all the frames while
people scroll or hold the screen, light enough to open fast on a
phone.

First ask me about my version: the house (style, materials, where it
stands), the phases I want to show, the words on the page and my
brand. Then follow the PDF in order. After every step show me the
result in the browser and wait for my OK before the next one. Use the
prompts in the PDF as they are, changing only the parts about my
house.
```

Claude is good at the code and the page, and it will write the 3D, the screenshots, the frame extraction and the page for you. The photos and the clips you make yourself on Higgsfield with the prompts in chapters 05 and 06, because that is where your eye matters most: you decide which photo is right before anything goes on.

> **Keep the chapters in order. Every step uses the result of the one before, and the most common mistake is jumping to the clips with phases that do not line up.**



CHAPTER 01

THE PHOTO THAT SETS THE HOUSE

Everything starts from one photo of the finished house, because that is the house the 3D copies and the place where the camera starts and ends.



A1 · EYE LEVEL, AFTERNOON



A2 · THREE METRES UP, MORNING MIST, THE ONE WE CHOSE

We made two versions and chose the one from slightly above, because it shows the whole lot with room around the house, so the camera can circle it without the roof going out of frame, and the morning mist between the trees gives every phase the same soft light. Take your photo the same way: the whole house, a wide lawn in front, the forest close behind and taller than the roof.

PROMPT 01 · NANO BANANA PRO · 16:9 · THE HOUSE

Architectural photograph taken from a slightly raised position, about three metres above the ground, with a 28mm lens: a three-quarter view of a finished two-storey American colonial house standing in a wide clearing deep in a tall American forest, with very tall oaks, maples and white pines surrounding the clearing on every side and rising well above the roofline behind the house. Timber-framed house with white horizontal lap siding, a steep dark charcoal shingle roof with gables and dormers, a tall red brick chimney on the left gable, white-trimmed windows with dark shutters, a white covered front porch, a garage wing on the right. Large open front lawn in the foreground, freshly finished: lush green grass with mowing stripes, a curved gravel driveway, boxwood hedges, hydrangeas and young dogwoods along the house. Soft early morning sunlight coming low through the trees from the right, light mist between the trunks, clear pale sky. Hyperrealistic photograph, natural colors, real texture of siding, shingles, brick and grass. No people, no cars, no text.

Write the materials in plain words (white lap siding, dark charcoal shingles, red brick chimney) because the 3D and every phase will be checked against this list.



CHAPTER 02

THE HOUSE
IN CODE

Claude copies the photo in three.js, in real metres, so the 3D can give every phase the same camera.

PROMPT 02 · CLAUDE CODE · THE 3D

Copy the house in this photo in three.js, in real metres: a two-storey American colonial with white horizontal lap siding, a steep dark shingle roof with the central gable and two dormers, the brick chimney on the left, the white porch with columns, the garage wing on the right, in a clearing of tall oaks and pines.

Make the siding out of real boards, about 18 cm each, and give every material a texture measured in metres, so nothing looks stretched when the camera gets close.

Put the camera where the photo was taken from, about three metres up with a 28mm lens, then make it go all the way around the house from left to right, one stop every 22.5 degrees, and save a 16:9 screenshot at every stop.

Put a ring of tall trees all around the clearing, beyond the path of the camera, and a deeper forest behind them, so every screenshot has trees behind the house and never an empty horizon.



THE 3D FROM THE PHOTO'S POINT OF VIEW



THE SAME 3D, ANOTHER STOP ON THE RAIL

The first version is never right, so the way to correct it is the same every time: put the 3D next to the photo, frame against frame, and ask Claude for exactly the difference, like the chimney too thin, the dormers too far apart or the porch too low. Our house is a white colonial with the chimney on the left gable, a covered porch with columns and a garage wing on the right, inside a ring of 120 trees between 50 and 74 metres from the centre and a deeper forest of 1,600 trees behind them.

> **The 3D does not need to be pretty. It needs the right shapes in the right places, because Nano Banana Pro will dress it later: the camera and the lines come from the 3D, the look comes from the photo.**



CHAPTER 03

THE RAIL AND THE SCREENSHOTS

One camera path around the house, one screenshot every 22.5 degrees, and the same lens and height on all of them.

CODE · THE CAMERA RAIL AROUND THE HOUSE

```
const C = [-3.5, 0, 0.5];                  // the centre of the orbit,
in metres
const at = (az, R = 36, y = 5.2, ty = 4.2) => {
  const a = az * Math.PI / 180;            // degrees from the street,
turning left
  return { p: [C[0] + R * Math.sin(a), y, C[2] - R * Math.cos(a)],
           t: [C[0], ty, C[2]] };          // camera 5.2 m up, looking
at 4.2 m
};
const A2 = 20;                             // where photo A2 was taken
from
const STOPS = Array.from({ length: 17 }, (_, i) => A2 - i * 22.5);
```

The camera sits 36 metres from the centre of the house, 5.2 metres up, looking at a point 4.2 metres high, and it starts exactly where photo A2 was taken. It goes all the way around from left to right in seventeen stops, and every stop gives a 16:9 screenshot with a 30mm lens. From the sixteen screenshots we kept nine, the ones where the house reads best, and each one became a phase.





Turn always in the same direction. When the camera turns right, the trees behind the house slide from left to right, and the clips in chapter 06 have to keep that direction or the move breaks.



CHAPTER 04

THE PHASES, BACKWARDS

Start from the finished house and take it apart, so that played forward the house rises while the camera turns.

We gave every screenshot a stage of the build, in the order a real house goes up, and the first frame of all is a blueprint of the same house on a desk under a brass lamp, made from the first screenshot and the photo. These are the names that also appear on the page, one per phase.



01 STAKED ON

the cleared lot only, wooden stakes and string marking the footprint, and the outline of the house drawn in thin glowing lines of light where it will stand.



02 FOUNDATION

the poured concrete foundation and slab, cured and clean, nothing built above it yet.



03 THE WALLS GO UP

the floor deck down and the first-floor stud walls standing in fresh yellow pine, the second floor just started.



04 FULLY FRAMED

the full timber frame up on the foundation, second floor, rafters, roof trusses and dormers in fresh yellow pine, no sheathing yet.



05 CLOSED IN

every wall and the whole roof covered in OSB structural panels, window and door openings cut, no wrap yet.



06 DRIED IN

white house wrap on every wall, the windows set, the roof shingled in dark charcoal.



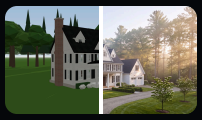
07 BRICK BY BRICK

the red brick chimney going up on the left gable and a pallet of bricks on the dirt, the rest of the house dried in.



08 EXTERIOR COMPLETE

white lap siding, trim and dark shutters finished, porch columns in, the lot still bare dirt.



09 FINAL TOUCHES

the finished house with a green lawn, a curved gravel drive, boxwood hedges, hydrangeas and young dogwoods, the dirt gone.



CHAPTER 05

NINE PHOTOS,
CAMERA LOCKED

Each 3D screenshot becomes the photo of one phase, with the camera exactly where the 3D put it.

PROMPT 05 · NANO BANANA PRO · 16:9 · ONE PHASE

```
Image 1 is a 3D screenshot. CAMERA LOCK: keep exactly the camera, the
framing, the lens and every line of the building from Image 1, do not
move anything.

Turn it into a real photograph of the same house at this stage: [the
stage line from the table]

Same clearing in a tall American forest, bare dirt with tyre tracks
all around, soft early morning sun coming low through the trees from
the right, light mist between the trunks. Hyperrealistic, natural
colours. No people, no machines, no text.
```

THE STAGE LINES

01 **Staked on.** the cleared lot only, wooden stakes and string marking the footprint, and the outline of the house drawn in thin glowing lines of light where it will stand.

02 **Foundation.** the poured concrete foundation and slab, cured and clean, nothing built above it yet.

03 **The walls go up.** the floor deck down and the first-floor stud walls standing in fresh yellow pine, the second floor just started.

04 **Fully framed.** the full timber frame up on the foundation, second floor, rafters, roof trusses and dormers in fresh yellow pine, no sheathing yet.

05 **Closed in.** every wall and the whole roof covered in OSB structural panels, window and door openings cut, no wrap yet.

06 **Dried in.** white house wrap on every wall, the windows set, the roof shingled in dark charcoal.

07 **Brick by brick.** the red brick chimney going up on the left gable and a pallet of bricks on the dirt, the rest of the house dried in.

08 **Exterior complete.** white lap siding, trim and dark shutters finished, porch columns in, the lot still bare dirt.

09 **Final touches.** the finished house with a green lawn, a curved gravel drive, boxwood hedges, hydrangeas and young dogwoods, the dirt gone.

> **The rule that made it work:** **the 3D screenshot goes in as Image 1 and nothing else. When we added the photo of the finished house as a second reference, the camera jumped back to that photo every time, so phases 02, 03, 04, 06, 08 and 09 were made again with the screenshot alone and then they lined up.**

Before you go on, put the nine photos in a row and look only at the camera: the horizon at the same height, the same lens, the trees in the same places. One photo out of line costs you a clip later.



CHAPTER 06

NINE CLIPS, START AND END

Every photo is the end of one clip and the start of the next, so nine clips played in a row look like one single move.

PROMPT 06 · KLING 4.0 · 1080P · 5 S · ONE CLIP

```
Image 1 is the start frame and image 2 is the end frame. The camera
keeps orbiting slowly to the right around the house at the same
height and the same distance, one smooth continuous move with no
cuts, while the build goes from [phase A] to [phase B]. The trees
behind the house slide from left to right. Same morning light and
mist from start to end. No people, no machines, no text.
```



START · CLOSED IN



END · DRIED IN

We made the clips on Kling 4.0 at 1080p and 5 seconds, 16.25 credits each, always with the start and the end frame. After every clip we pulled

its first and last frame and compared them with the two photos, and when the difference was visible we made the clip again: clips 3, 7 and 9 were redone, clip 3 because the trees turned the wrong way.

THROUGH THE FRONT DOOR



THE END FRAME OF THE WALK INSIDE

To walk into the house we recorded the 3D going from the lawn through the door, cut it to exactly 5 seconds, and gave it to the Kling 4.0 video edit with the last outside photo as start and the wide photo of the living room as end, 40 credits a video. The second video, from the kitchen to the living room, starts on the real last frame of the first one, taken from the video file, never on a photo that looks close.

PROMPT 06B · KLING 4.0 VIDEO EDIT · THE WALK INSIDE

```
Follow the camera of the video exactly: from the lawn to the front
door, through the door and into the house, one continuous move. The
start frame is the first frame, the end frame is the wide photo of
the living room. Keep the house, the light and every material of the
two photos. No people, no text.
```

> **Always set the duration to 5 seconds and always give the end frame. Without the end frame the edit invents its own room, and the credits are gone.**



CHAPTER 07

FRAMES LIGHT ENOUGH FOR A PHONE

The page does not play a video. It draws one frame at a time, and the frames have to arrive fast on a phone.

CODE · CUT THE CLIPS INTO FRAMES

```
# one folder of frames from the nine clips, in order
for i in 1..9:
    ffmpeg -i clip0{i}_*.mp4 -vf "scale=-2:1080,crop=1920:1080" -q:v
3 c{i}/%04d.jpg
    from clip 2 on, drop the first frame: it is the last frame of the
clip before
    move the frames into frames/0000.jpg, 0001.jpg ... and note where
each clip ends
```

The nine clips and the two walks inside give 960 frames in a row. Saved as JPG they weigh 208 MB, so on a phone the build was still loading after a minute, and the fix was to turn every frame into AVIF: 1280 pixels wide for the PC and a vertical cut of 540 by 960 for the phone, from the original frames, never from a smaller copy.

TERMINAL · JPG TO AVIF

```
# PC: 1280 px, every frame
ffmpeg -i frames/0500.jpg -vf scale=1280:-2 -c:v libaom-av1 -still-
picture 1 -crf 33 -cpu-used 6 d/0500.avif
# phone: a vertical cut 540x960
ffmpeg -i frames/0500.jpg -vf "crop=608:1080:656:0,scale=540:960" -
c:v libaom-av1 -still-picture 1 -crf 32 -cpu-used 6 m/0250.avif
```

	BEFORE	AFTER
Weight	208 MB	PC 37.6 MB · phone 8.4 MB
Full build on a phone (4G)	over 60 s	10 s

CODE · THE LOADING ORDER

```
const order = [], seen = new Set();
const push = i => { if (MOB) i -= i % 2; if (i >= 0 && i < N &&
!seen.has(i)) { seen.add(i); order.push(i); } };
META.bounds.forEach(push);                          // first the end of
every phase
for (const step of [24, 12, 6, 3, 2, 1])            // then denser and
denser passes
  for (let i = 0; i < N; i += step) push(i);
```

The first frame is also the poster in the HTML and loads first, then the end of every phase, then passes that get denser and denser, so the

page can start after a few seconds and fill in while people watch. Phones take one frame in two, and nobody sees the difference.



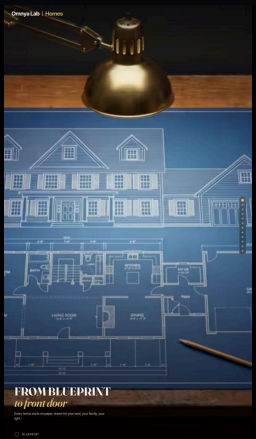
CHAPTER 08

THE HOLD TO BUILD PAGE

One canvas fixed on the screen, a long empty scroll behind it, and the frame that follows the scroll.



THE PAGE ON A PC



ON A PHONE

CODE · SCROLL, FRAME, DRAW

```
const PX_PER_FRAME = 9;                          // how long the
build lasts on the scroll
function progress() {                            // 0 at the top of
the build, 1 at the end
  const r = build.getBoundingClientRect();
  return Math.min(1, Math.max(0, -r.top / (N * PX_PER_FRAME)));
}
function nearest(i) {                            // never wait: show
the closest frame already loaded
  if (ok[i]) return i;
  for (let d = 1; d < N; d++) { if (ok[i - d]) return i - d; if (ok[i
+ d]) return i + d; }
  return -1;
}
function loop() {                                // the frame
follows the scroll softly
  cur += (target - cur) * 0.12;
  draw(cur);
  requestAnimationFrame(loop);
}
```

Every frame takes 9 pixels of scroll, so the whole build lasts about 8,600 pixels, and the frame never jumps to the scroll but follows it softly, which is what makes it feel heavy and real. The picture zooms in by 3.5 percent over the whole build, so even a slow scroll keeps moving, and if a frame has not arrived yet the page shows the closest one it has instead of waiting.

On top of the frames, every phase gets its line of words, its degrees around the house and a dot on the side rail, all taken from the same list: The plan, Site 0°, Foundation 67°, Framing 112° and 157°, Sheathing 202°, Dried in 247°, Masonry 292°, Exterior 315°, Landscape 360°, then Step inside and the living room. At the end the picture blurs and the logo comes in.

> **On a phone people hold the screen instead of scrolling: the page turns that into the same slow scroll from the first frame to the last in 34 seconds, and stops the moment they touch it.**



CHAPTER 09

DELIVERY AND WHAT TO CHARGE

The client gets one link that opens fast on every phone, and you keep the files.

Every project is a static folder, an index.html and its frames, with no build step. We put each client on their own Vercel project with their own subdomain, a CNAME record like 3d.client.com, so one client never weighs on another and the link keeps working as long as they keep the hosting.

IF THEY WANT IT INSIDE THEIR SITE

CODE · THE IFRAME TO PASTE IN THEIR SITE

```
<iframe src="https://3d.client.com/" title="Hold to build"
  style="width:100%;height:100vh;border:0;display:block"
  allow="fullscreen; autoplay" loading="lazy"></iframe>
```

A build that plays on the scroll needs the iframe as tall as the screen, so the scroll lives inside it. When the client has a full page free, a link to the full page is always better.

THE LAST CHECK

Open the online link in PageSpeed, mobile mode, before you send it: the first picture has to be on screen in under 2.5 seconds. Our house scores 96 to 99 on a phone.

WHAT WE CHARGE

- An interactive 3D like this one starts at **€1,990** on our price list
- **Hosting and updates** at €49 a month keeps the link online and changes the words or the phases when they ask
- An **offline version** for a sales office, full quality on a screen with no Wi-Fi, is €490 more and only on request

> **Want us to build it for your house? Send the plans or a photo to info@omnyalab.com, the first image is free.**